

Moumita Haque Mohona

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PROFESSIONAL PROFILE

B.Sc. (Engineering) graduate in Information and Communication Engineering from Pabna University of Science and Technology with a CGPA of 3.90/4.00, ranked 1st in the class. Research-focused in Artificial Intelligence, Deep Learning, Computer Vision, Biomedical Imaging, Signal Processing, and Explainable AI, with published works in IEEE and Springer platforms. Experienced in developing AI-based solutions for healthcare and intelligent systems.

EDUCATION

Bachelor of Science in Engineering (B.Sc. Engg.) in Information and Communication Engineering | *Pabna University of Science and Technology (PUST)* 2026

- Undergraduate Thesis: "A YOLO-Comparative Multimodal Framework for Object, Emotion, and Activity-Aware Bilingual Image and Video Captioning"

Higher Secondary Certificate (HSC) – Science | *Kumudini Govt. College, Tangail* 2020

- GPA: 5.00/5.00

Secondary School Certificate (SSC) – Science | *Police Lines Adarsha High School, Tangail* 2018

- GPA: 5.00/5.00

ACADEMIC DISTINCTION

Academic Excellence Award – Department of Information and Communication Engineering, PUST (2025) — Received the Academic Excellence Award in recognition of outstanding academic performance, dedication, and commitment to academic excellence during the undergraduate program.

Academic Excellence Award – ICE Fiesta 2025, PUST — Honored by the Department of Information and Communication Engineering, Pabna University of Science and Technology, for outstanding academic achievement and excellence during the academic year 2025.

PUBLICATIONS & RESEARCH DEVELOPMENT

- Remote Health Monitoring via PPG Signal Processing and Machine Learning Using Mobile Devices.** 2025 28th International Conference on Computer and Information Technology (ICCIT), IEEE, 2026.
- Machine Learning-Based Hepatitis C Detection Using Ensemble Boosting.** *Lecture Notes in Networks and Systems (LNNS)*, Springer Nature, 2026.
- Machine Learning-Based Prediction of Nightmares in Children: A Pediatric Psychological Integration.** 2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS), IEEE, 2026.
- GR-ACE Net: A Hybrid Graph-Attentional Framework with Global Relational Reasoning for Deepfake Forensics.** 2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII), IEEE.
- A Comparative Study of CNN, Transformer, and Recurrent Models for Deepfake Detection on Degraded Images.** 2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII), IEEE.
- A Hybrid YOLO-EfficientNet Framework for Fine-Grained Vegetable Recognition in Dense Multi-Object Images.** 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), IEEE.

Awards & Achievements

- Research Paper Presentation Recognition, IEEE ICCIT 2025 : Presented "Remote Health Monitoring via PPG Signal Processing and Machine Learning Using Mobile Devices."
- Research Paper Presentation Recognition, IEEE COMPAS 2025 : Presented "Machine Learning-Based Prediction of Nightmares in Children: A Pediatric Psychological Integration."
- Research Paper Presentation Recognition, BIM 2025 (Springer/Taylor & Francis supported) – : Presented "Machine Learning-Based Hepatitis C Detection Using Ensemble Boosting."
- Research Paper Presentation Recognition, IEEE QPAIN 2026 – : Presented research on "A Hybrid YOLO-EfficientNet Framework for Fine-Grained Vegetable Recognition in Dense Multi-Object Images."

TECHNICAL & RESEARCH SKILLS

Programming: Python, C, C++, Java, JavaScript, PHP, SQL, MATLAB

AI & Computer Vision: PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas, Matplotlib

Research & Data: Field image acquisition, bounding-box annotation, dataset curation and splitting, model training and evaluation, literature review, scholarly writing

Systems & Tools: Jupyter Notebook, Kaggle, Visual Studio Code, SQL Server Management Studio, Proteus, mikroC, Cisco Packet Tracer

SELECTED ACADEMIC COURSEWORK

Artificial Intelligence and Robotics • Neural Networks • Natural Language Processing • Digital Image and Speech Processing • Digital Signal Processing • DBMS • Web Programming • Software Analysis and System Testing • Wireless Communications • Computer Architecture and Microcontroller Design

PROJECTS, CERTIFICATIONS & LEADERSHIP

Projects & Certification: SQL (Basic) - HackerRank; Universal Control Board System (Microcontroller and hardware interfacing).

REFERENCES

Dr. Md. Anwar Hossain

Professor, Department of Information and Communication Engineering, PUST

Thesis Supervisor

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Dr. Md. Sarwar Hossain

Professor, Department of Information and Communication Engineering, PUST

Academic Referee

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